18.C21 Midterm
Spring 2026 (Johnson & Marzouk)

NAME: .....

- You have **50 minutes**. **Please allocate your time carefully**, to make sure you get to work on all parts—take a quick look at the *whole exam* before starting.
- Write your **NAME** above *and* at the top of *every* page. Please do it as soon as you begin the exam, and check again before returning it.
- Only write on *one* side of each sheet. Additional paper will be provided upon request.
- You can bring **1 letter-sized sheet of notes** (both sides if you want).
- **Show your work / justify your answers.** Wrong answers may still receive partial credit *if* the reasoning was somewhat correct.
- If you absolutely believe there is something wrong with a problem, please ask. No hints, other than the ones already written in the exam, will be given.

A few useful formulas and facts from class

- Double-precision (`Float64`) floating-point numbers store about 15–16 decimal significant digits, and can represent magnitudes from around $\approx 10^{-300}$ to around $\approx 10^{300}$ (after which they under/overflow, respectively).
- A least-square problem $\min_x \|Ax - b\|_2$ is minimized when $A^T Ax = A^T b$.
- The Chebyshev points for polynomial interpolation on $[-1, 1]$ are $x_k = \cos(k\pi/n)$ for $k = 0, \ldots, n$.
- A Newton step for $f(x) = 0$ is $x_{k+1} = x_k - f'(x_k)^{-1} f(x_k)$.
- Piecewise-linear interpolation with intervals Δx has an error $O(\Delta x^2)$ for twice-differentiable functions.
- Polynomial interpolation from Chebyshev points, and resulting algorithms like Clenshaw–Curtis quadrature, converges exponentially fast for any function with a Taylor series (or at least faster than any power law if the function is infinitely differentiable).

Problem 1 (10+10 points)

The roots x_1, x_2 of a quadratic equation $ax^2 + bx + c = 0$ can be found from the quadratic formula, but these solutions can be written in a different form by instead solving $a + bx^{-1} + cx^{-2} = 0$ for x^{-1}, resulting in two possible expressions for each root:

$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} = \frac{2c}{-b - \sqrt{b^2 - 4ac}},$$
$$x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a} = \frac{2c}{-b + \sqrt{b^2 - 4ac}}.$$

Call the formula on the left the "regular" formula and the one on the right the "inverted" formula.

(a) For a particular choice of a, b, c, you find that the "inverted" formula for x_1 is *much* more accurate than the "regular" formula in double/`Float64` precision. Give example values of a, b, c where that would be true. Also, which of the two formulas for x_2 will be more accurate in this case?

(b) You find in `Float64` precision that the regular formula for x_1 is giving `Inf` and the inverted formula for x_1 is giving `0.0`, but none of a, b, c are `Inf` or `0.0`, and an arbitrary precision calculation tells you that the correct answer for x_1 should be nonzero and representable as a finite value in `Float64` precision. Give example values of a, b, c where this might be happening, and explain how to correct the formulas to handle this case accurately while staying in `Float64` precision.

Problem 2 (10 points)

Suppose that you are given a function value $f(x_1)$ and $f(x_2)$ at two points x_1, x_2 and are also given the *derivative* $f'(x_1)$ and $f'(x_2)$ at those two points.

Explain how to find a polynomial interpolant $p(x)$ that matches *both* f and f' at x_1, x_2. What should the degree of $p(x)$ be? Write down a system of linear equations for the coefficients of p. (You don't need to solve them.)

Problem 3 (10+10 points)

Suppose that you are trying to compute

$$I = \int_0^2 |f(x)|\, dx\,.$$

where

$$f(x) = \sin(x - \cos(x))\,,$$

and the integrand $|f(x)|$ is plotted in Fig. 1 below along with $f(x)$.

(a) If you estimated this integral using the trapezoidal rule with $n+1$ equispaced points (n intervals), what would you expect the convergence rate to be? For example, is the error $O(1/n^2), O(1/n^{1.5}), O(1/n), \ldots$? Justify your answer (briefly).

(b) Outline an algorithm to get I with exponentially fast convergence using Clenshaw–Curtis quadrature and *some other ingredient*. (Assume you have a subroutine I_n = `clencurt(func,a,b,n)` that estimates the integral $\int_a^b \text{func}(x)dx$ of any given function `func` by Clenshaw–Curtis quadrature on any given interval $[a, b]$ and for any number of points $n+1$; you need not outline the Clenshaw–Curtis algorithm itself.) Why is Clenshaw–Curtis *alone* not enough here?

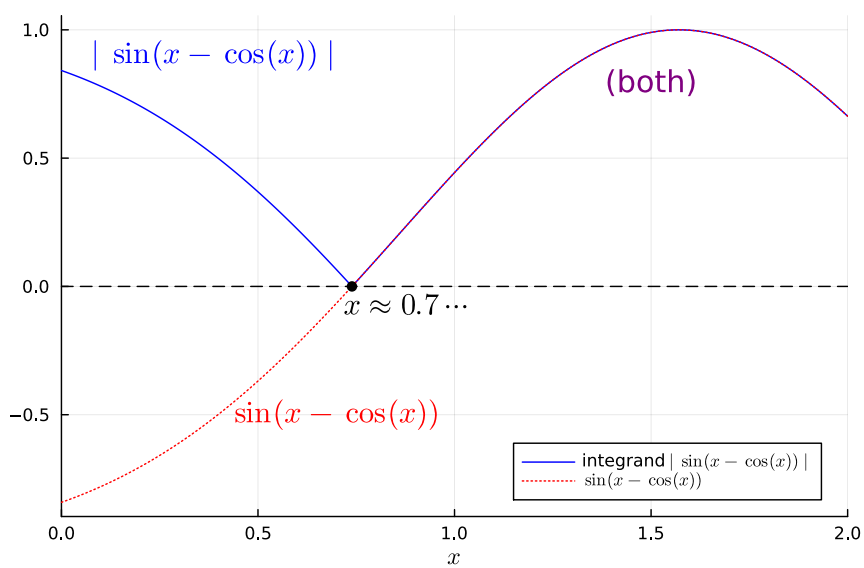

Figure 1: Plot of the integrand $|f(x)| = |\sin(x - \cos(x))|$ (blue line) along with the function $f(x) = \sin(x - \cos(x))$ without the absolute value (red dotted). The zero crossing (black dot) occurs at an irrational x that is $\approx 0.7\cdots$.